

VBA Cheat Sheet for Analyst

VBA Data Types

Variant	any data type
Integer	(2 bytes) integer
Long	(4 bytes) integer
Single	(4 bytes) floating point
Double	(8 bytes) floating point
String	non-numeric data (declared with quotes)
Object	any object reference
Date	a date
Boolean	True / False
Byte	0-255

Operators (Syntax)

Comparison	=, <> (not equal to), >, <, >=, <=
Logical (Boolean)	NOT (opposite), AND (if all <i>true</i> then returns true), OR (at least 1 <i>true</i> returns true)
Mathematical	+, -, *, /, \ (integer division), Mod (remainder), ^ (remember to put space)
String Concatenation	&

The concatenate operator eg. "A" & "B" becomes "AB".

Operators (Precedence)

1	^
2	* OR / (division)
3	\ (integer division)
4	Mod
5	+ OR -

Declarations

Variables	Dim [varname] As [type]
Arrays	Dim [arrayname(index)] As [type]
Re-declare Array	ReDim [arrayname(newindex?)] As [newtype?]

Declarations (cont)

ReDim Preserve [arrayname(newindex?)] **As** [newtype?]
and
keep
values

Option Explicit is used to require declarations of all variables. **Option Base 1** makes the index of all arrays starts from 1. If declared array has no numeral index provided, then it is dynamic in size.

Data Functions (Conversions)

...value to a boolean	CBool (value)
...value to a integer	CInt (value)
...value to a double	Cdbl (value)
...value to a string	CStr (value)

Val function accepts a string as input and returns the numbers found in that string.

Math Functions

Absolute	Abs ([numeric value])
Square root	Sqr ([numeric value])
Exponential, e	Exp ([numeric value])
Natural log, ln	Log ([numeric value])
Is it a number (boolean)?	IsNumeric ([numeric value])
Truncate to integer	Int ([numeric value])
(Num1 / Num2) remainder?	[Num2] Mod [Num2]
Round to a decimal place	Round ([numeric value], [# of digits])

[**Sin**/**Cos**/**Tan**] (x) for trigonometric functions,
[**ASin**/**ACos**/**ATan**] (x) for inverse trig functions.

String Functions

All upper case	UCase ([string value])
All lower case	LCase ([string value])
Length of string (integer)	Len ([string value])
Filters a string to a double	Val ([string value])
Convert number to string	Str ([numeric value])

Val function accepts a string as input and returns the numbers found in that string.

User Interaction and Cell Selection	
A popup dialogue box	MsgBox "dialogue" [& variable etc.]
Prompt user for input	InputBox ("dialogue")
...a single cell A1	Range ("A1").Select
...an active cell	ActiveCell .Select
...a contiguous range	Range ("A1:G5").Select
...offset and select	[ActiveCell/Range(#)].Offset (1,0).Select
...set a cell's value	[ActiveCell/Range(#)].Value = [varname]
Use Range to select specific cells or group of cells. Use ActiveCell to select highlighted cell in excel.	

Array Functions	
Highest element number	UBound (ArrayName [, Dimension])
Lowest element number	LBound (ArrayName [, Dimension])
Highest element number is the size of possible entries a array can hold.	

Array Iteration	
<pre>Function MinIntegerofArray(TheArray As Variant) As Integer Dim i As Integer, placeholder As Integer placeholder = 0 For i = 1 To UBound(TheArray) If TheArray(i) < TheArray(placeholder) Then placeholder = i End If Next 'index of min value is at placeholder MinIntegerofArray = TheArray(placeholder) End Function</pre>	

Error Handling with GoTo & Labels	
<pre>... tempstudentName = InputBox("Please enter student name (type exit to end): ") 'error check and force reentry of input If IsText(tempstudentName) = False Then checker1: tempstudentName = InputBox("Please enter a valid student name [not blank and letter] (type exit to end): ") End If If tempstudentName = "exit" Then End '(the program) ElseIf IsText(tempstudentName) = False Then GoTo checker1 'label Else studentName = tempstudentName End If ... 'checker1:' is a label: labels only include the next line</pre>	

IsText, Case Statements, For loop	
<pre>Function IsText(streng As Variant) As Boolean Dim i As Integer For i = 1 To Len(streng) 'checks if the text follows the ACII numerals (a- z AND A-Z) Select Case Asc(Mid(streng, i, 1)) Case 65 To 90, 97 To 122 IsText = True Case Else IsText = False Exit For End Select Next End Function</pre>	